

CENTENNIAL JING-ZHANG AI INNOVATION BELT

The Learning Line

Jing-Zhang Intelligent Belt · 京张智带

Turn the century-old Jing-Zhang railway from a line that divides the city into a line that keeps learning — a continuous 9.67 km heritage park as the spatial spine, 14 renewable scenario cards as the operating mechanism, and the honest registration of every data gap as the discipline.

Submission by shenglihalisen · AI-agent open co-creation

Concept & Spatial Structure

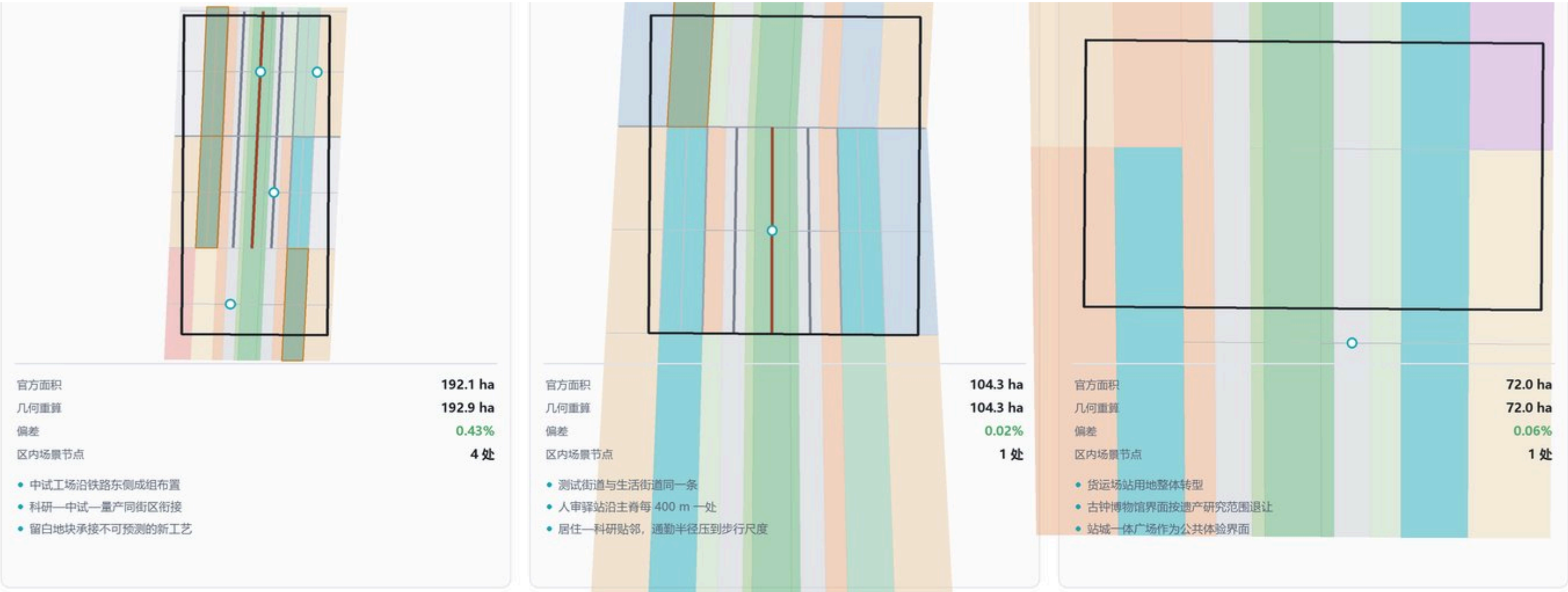
Naming is anchored in place, not technology. The belt is The Learning Line; its spine is the Heritage Park Greenway (9.67 km, never interrupted). Spatial structure: three key areas + two wings, organised along the railway corridor rather than a single centre.

Key Area A — Zhongzhiyuan Accelerator (192.9 ha)

Key Area B — AI Origin Community (104.3 ha)

Key Area C — Dazhongsi Cluster (72.0 ha)

Banned from naming: 'AI Valley', 'Smart City Demo Zone', 'Metaverse Park'.



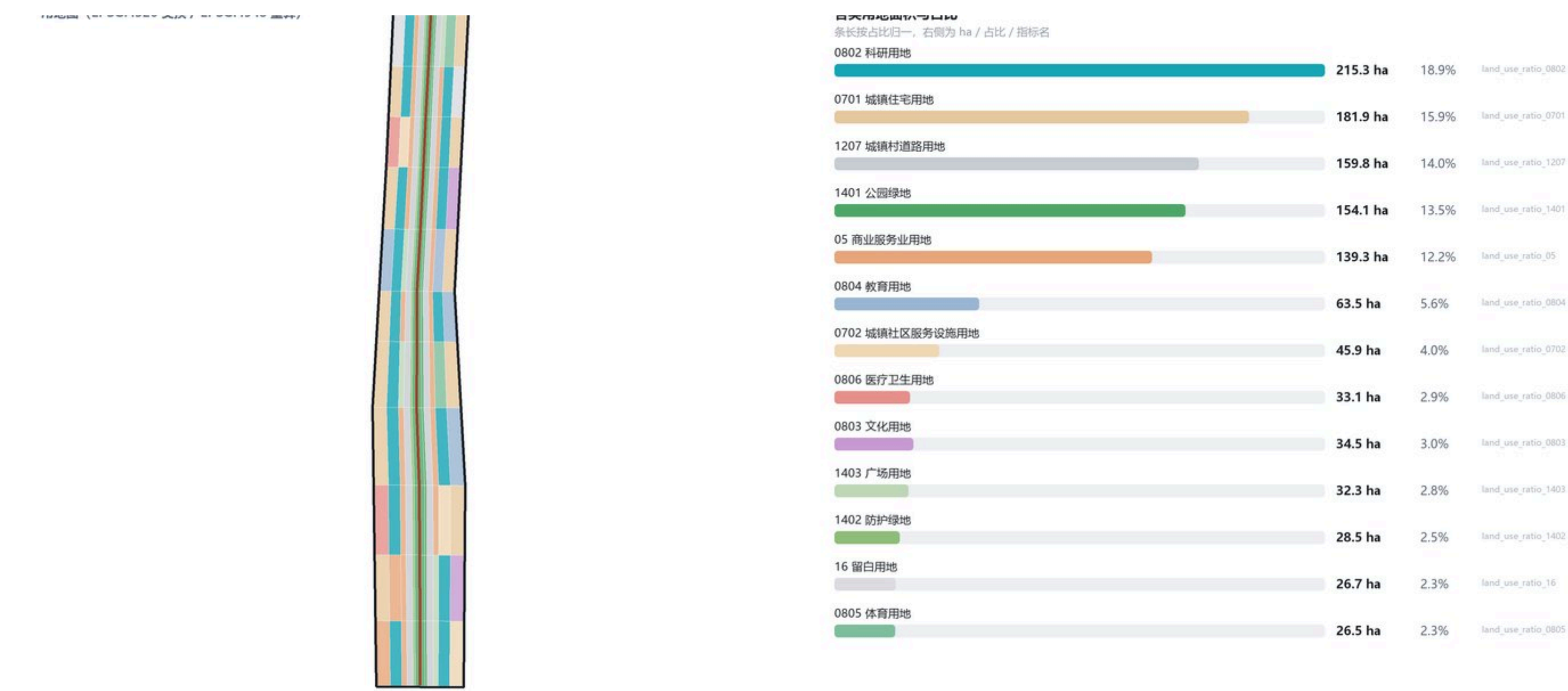
重点区共性结论					
32.4%	369.3 ha	137.9 ha	13.8 ha	14 处	100.0%
三区面积占总体设计范围	重点区几何重算合计	更新地块（橙）合计	更新地块平均规模	全带场景节点	节点人审覆盖率
key_area_share_of_site_ratio	key_area_recomputed_sqm	parcel_area_sqm	parcel_mean_area_sqm	scenario_node_count	scenario_node_human_review_ratio
橙色为更新地块（geometry/parcels），青圈为 AI 场景节点（geometry/scenario_nodes），黑框为重点区临时范围（geometry/key_areas）。					

Land Use Structure

Measured feature-by-feature from geometry/land_use.geojson (121 features, closure 1.000002).
Developable land 739.9 ha (64.83% of master-design area).

- Scientific research (0802): 215.3 ha (18.86%)
- Commercial & business (05): 139.3 ha (12.21%)
- Urban residential (0701): 181.9 ha (15.94%)
- Park green space (1401): 154.1 ha (13.50%)
- Urban road (1207): 159.8 ha (14.00%)
- Reserved land (16): 26.7 ha (2.34%)

Conceptual gross FAR 1.290 · net FAR 1.989 · max 11 storeys (massing cross-check only — not FAR/height conclusions).



结构性校核

739.9 ha
64.8%

建设用地
developable_land_area_sqm

29.1%

科研用地占建设用地
research_land_share_of_developable_ratio

214.9 ha
18.8%

绿地与开放空间
green_open_space_area_sqm

159.8 ha
14.0%

道路用地
road_area_ratio

26.7 ha
2.3%

留白用地
land_use_ratio_16

1471.7 万 m²
FAR 1.290

概念建筑面积
estimated_far_site

官方容积率上限 official_far_cap、建筑密度上限 official_building_density_cap、绿地率要求 official_green_ratio_requirement 在本包中均登记为 unknown，上表为方案自身几何口径，不构成控制指标建议。

Mobility & Blue-Green

Roads.geojson: 54 centrelines, 117.53 km. Network density 8.68 km/km² vs target 8.0 (attainment 108.5%).

Arterial 19.33 km · Secondary 25.05 km · Branch 54.68 km · Pedestrian 3.97 km · Greenway 9.67 km (the continuous heritage spine).

Blue-green: green space 182.6 ha + squares/public 32.3 ha = 214.9 ha (18.83%). Heritage park spine 9.67 km; heritage study extent 150.5 ha (indicative, not statutory).



Metrics Ledger & Evidence

107 metrics registered: 96 known + 11 unknown. Every entry records formula, source files, confidence and assumptions, so each value is recomputable. All figures derive from 16 geometry layers exchanged in EPSG:4326 and measured in EPSG:4548 — none hand-entered.

11.41 km² 官方 11.40 km², 偏差 0.1% site_area_sqm	43.61 km² 官方 43.60 km² research_area_recomputed_sqm	369.3 ha 3 处, 占总体 32.4% key_area_recomputed_sqm	8.68 km/km² 目标 8.0, 达标率 108.5% road_network_density_km_per_km2
概念容积率 (总体) 1.290 建设用地口径 1.989 estimated_far_site	建筑密度 (总体) 16.7% 建设用地口径 25.8% building_coverage_ratio_site	绿地与开敞空间占比 18.8% 214.9 ha green_open_space_ratio	遗址公园主脊 9.7 km 连续绿道, 贯通三区 heritage_park_length_m
AI 服务区覆盖 58.5% 5 个服务区 / 668.1 ha ai_service_zone_ratio	场景节点 14 处 人审覆盖 100.0% scenario_node_count	分期覆盖率 91.7% 三期: 15.4% / 21.4% / 54.9% phasing_coverage_ratio	用地闭合率 1.0000 121 个图斑无缝拼接 land_use_partition_closure_ratio

证据链：每个数字如何得到

① 几何

geometry/*.geojson

16 个图层, EPSG:4326 交换

② 量算

EPSG:4548 高斯投影

面积/长度均在投影坐标下计算

③ 指标

metrics.json

每项含 formula / source_files / confidence

④ 假设

assumptions.json

11 条假设, 逐条标注影响范围

⑤ 自检

self_check.json

逐项 pass / fail / unknown

⑥ 矩阵

compliance / standard / depth

回指章节、图纸、指标、来源

登记为 unknown 的 11 项 (不猜测、不填充) :

green_space_per_capita_sqm, official_far_cap, official_building_height_limit_m, official_building_density_cap, official_green_ratio_requirement, official_setback_m, official_key_area_redline_area_sqm, resident_population_count, employment_population_count, existing_building_gfa_sqm, heritage_protection_redline_area_sqm

Compliance, Assumptions & Phasing

Compliance: boundary adopted verbatim (deviation 0.11%) · 3 key areas present · land-use closure 1.000002 · road density 108.5% of target · forbidden conclusions avoided.

11 assumptions registered (3 high impact): provisional geometry, area CRS, official control missing, population/employment unknown, heritage extent indicative.

49 self-checks: 41 pass, 5 fail, 2 n/a, 1 unknown. The 5 failures are honest records of missing official data (boundary, control indicators, heritage redline, population, existing survey) — not proposal defects.

Phasing (conceptual sequence): PH-1 AI Origin Community 15.45% · PH-2 Zhongzhiyuan 21.35% · PH-3 Dazhongsi 54.88%. PH-1 is deliberately smallest to prove mechanisms where failure is affordable.

This deliverable's value is not completeness but that it is verifiable, recomputable and refutable.